Volatility Forecasting and Risk Signal Analysis

VIX Index — R

This report builds a simple forecasting and monitoring framework for the VIX index. The modelling work compares ARIMA and regression forecasts, then uses rolling thresholds to classify volatility conditions into Normal, Elevated and Stress regimes.

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1 Executive summary

This project forecasts the VIX index and converts the results into a simple market-risk signal. VIX is an implied volatility index, so the project is forecasting the index itself rather than realised volatility. The daily VIX sample runs from 1990-01-02 to 2026-05-21 and contains 9,191 clean observations. A rolling backtest was then run over 775 one-day-ahead forecasts.

The regression model had the lowest VIX-level RMSE at 1.7805, compared with 1.7865 for ARIMA. The difference is small, so the result should be read as a marginal improvement rather than a clear win. The interval coverage was close to the 95% target for both models: 94.58% for ARIMA and 94.84% for the regression model.

The residual checks gave a more cautious view. The ARIMA Ljung-Box p-value was 0.7791, while the regression p-value was 0.0106. This suggests the regression model was slightly better on point forecast accuracy, but the ARIMA residuals were cleaner from an autocorrelation perspective.

The risk signal splits the VIX history into three states using rolling thresholds based only on past data. Normal days made up 74.47% of the sample, Elevated days 12.73%, and Stress days 12.80%. Stress days had an average VIX of 27.93 and a maximum VIX of 82.69, which confirms that the signal captures high-volatility periods.

2 Data

The project uses daily closing values of the Cboe Volatility Index, commonly known as VIX. The data were downloaded from FRED using the VIXCLS series. FRED reports this as a daily, close, not seasonally adjusted index series sourced from Cboe [1]. Cboe describes the VIX index as a measure of expected 30-day volatility implied by S&P 500 option prices [2].

The data were cleaned by removing missing observations and retaining strictly positive VIX values. A log transformation was used for the forecasting models, since VIX is positive and can spike sharply during market stress.

Table 1: Data summary

Metric	Value
Data start	1990-01-02
Data end	2026-05-21
Clean observations	9191
Rolling forecasts	775
Risk signal observations	8939

3 Methodology

3.1 Forecasting models

Two forecasting models were used.

- **ARIMA model:** an automatic ARIMA specification was selected using an information criterion. This follows the standard approach implemented in the R `forecast` package, which is described by Hyndman and Khandakar [3].
- **Regression model:** a regression model was selected using AIC over lagged and rolling VIX features. The features included lagged log-VIX, short and medium moving averages, and previous risk-regime indicators. AIC is a standard model selection criterion introduced by Akaike [4].

The main evaluation used rolling one-day-ahead forecasts. This is more realistic than fitting a model once on the full sample, because the forecast at each date only uses information available before that date.

3.2 Risk signals

The risk signal is based on rolling VIX thresholds. For each date, the model compares the current VIX value with the 75th and 90th percentiles of the previous 252 trading days. This gives three regimes:

- **Normal:** VIX is below the elevated threshold.
- **Elevated:** VIX is above the 75th percentile threshold but below the stress threshold.
- **Stress:** VIX is above the 90th percentile threshold.

The thresholds are lagged, so the signal avoids look-ahead bias. This matters because the purpose is to produce a monitoring signal that could have been calculated at the time.

4 Forecasting results

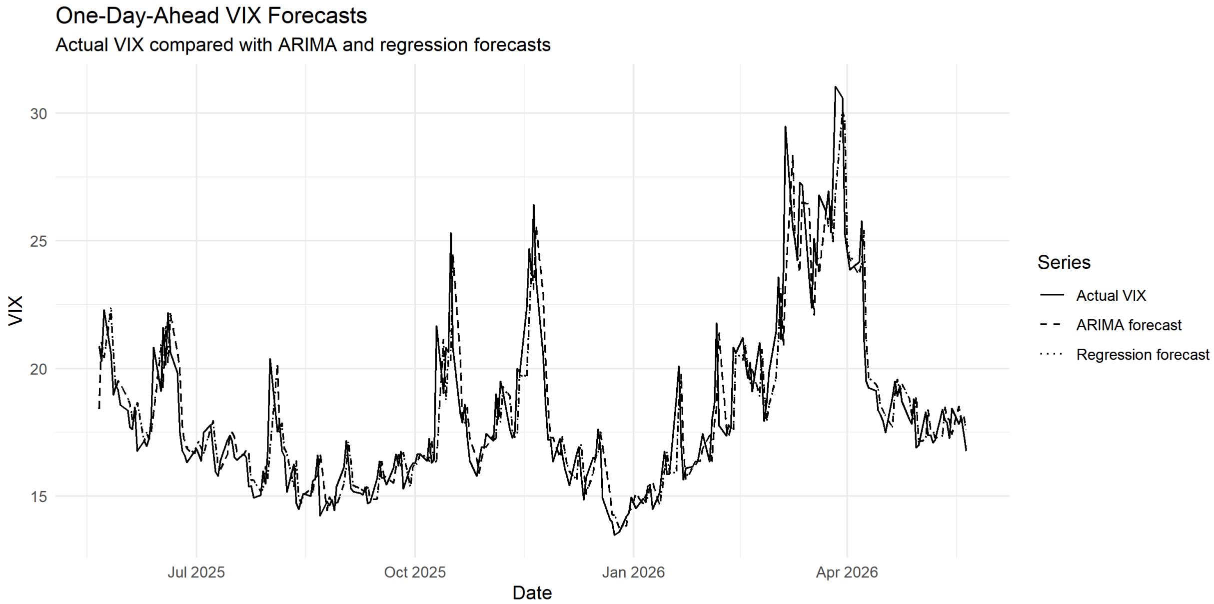

Figure 1: One-day-ahead VIX forecasts. The ARIMA and regression forecasts both track the broad movement in VIX, although sharp spikes remain difficult to anticipate.

The two forecasting models produced very similar point forecast accuracy. The regression model had a slightly lower RMSE on the VIX level, but ARIMA was very close. Both models also had 95% interval coverage close to the nominal target.

Table 2: Model comparison over 775 rolling one-day-ahead forecasts

Model	RMSE log	MAE log	RMSE VIX	MAE VIX	MAPE VIX (%)	95% cov. log (%)	95% cov. VIX (%)	Forecasts
ARIMA	0.0766	0.0514	1.79	0.9822	5.09	94.58	94.58	775
Regression	0.0765	0.052	1.78	0.9937	5.17	94.84	94.84	775

The small gap between the two RMSE values means the regression result should not be overstated. In practical terms, both models perform similarly on short-horizon VIX forecasts.

5 Forecast uncertainty

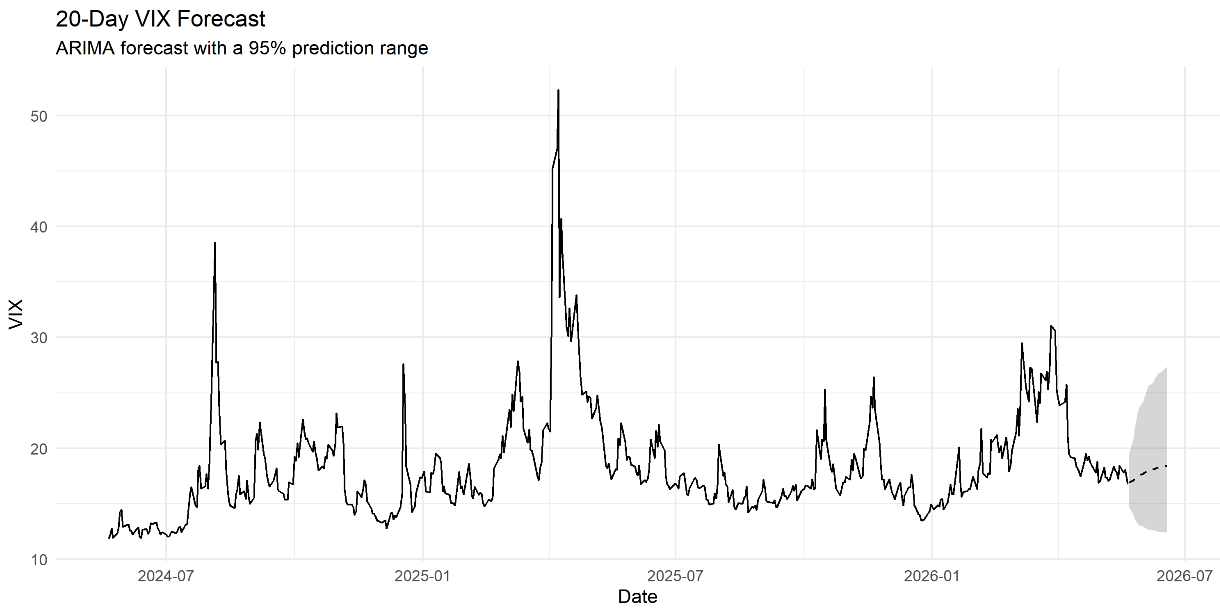

Figure 2: Final 20-day ARIMA forecast. The shaded region shows the 95% prediction range.

The final 20-day forecast is relatively flat, with a mild upward drift in the central forecast and a widening prediction range. This is expected: uncertainty grows as the forecast horizon extends. The interval is more informative than the point forecast because VIX can move sharply during stress periods.

Table 3: Final 20-day VIX forecast, selected columns

Date	Mean VIX	Lower 95%	Upper 95%
2026-05-22	16.94	14.66	19.57
2026-05-25	17.10	14.06	20.79
2026-05-26	17.25	13.68	21.74
2026-05-27	17.38	13.42	22.50
2026-05-28	17.50	13.23	23.15
2026-05-29	17.61	13.08	23.71
2026-06-01	17.71	12.96	24.20
2026-06-02	17.80	12.87	24.63
2026-06-03	17.89	12.79	25.00
2026-06-04	17.96	12.73	25.34
2026-06-05	18.03	12.68	25.64
2026-06-08	18.09	12.63	25.91
2026-06-09	18.15	12.59	26.16
2026-06-10	18.20	12.56	26.38
2026-06-11	18.25	12.52	26.59
2026-06-12	18.29	12.49	26.77
2026-06-15	18.33	12.47	26.94
2026-06-16	18.36	12.44	27.10
2026-06-17	18.39	12.42	27.24
2026-06-18	18.42	12.40	27.37

6 Residual checks

Residual diagnostics were used to check whether the models left behind strong autocorrelation. The Ljung-Box test is a standard portmanteau test for remaining autocorrelation in time-series residuals [5].

Table 4: Ljung-Box residual tests

Model	Test	Lag	Statistic	p-value	Interpretation
ARIMA	Ljung-Box	20	11.48	0.7791	No strong evidence of residual autocorrelation at 5% level.
Regression	Ljung-Box	20	28.95	0.0106	Residual autocorrelation remains; model may miss time-series structure.

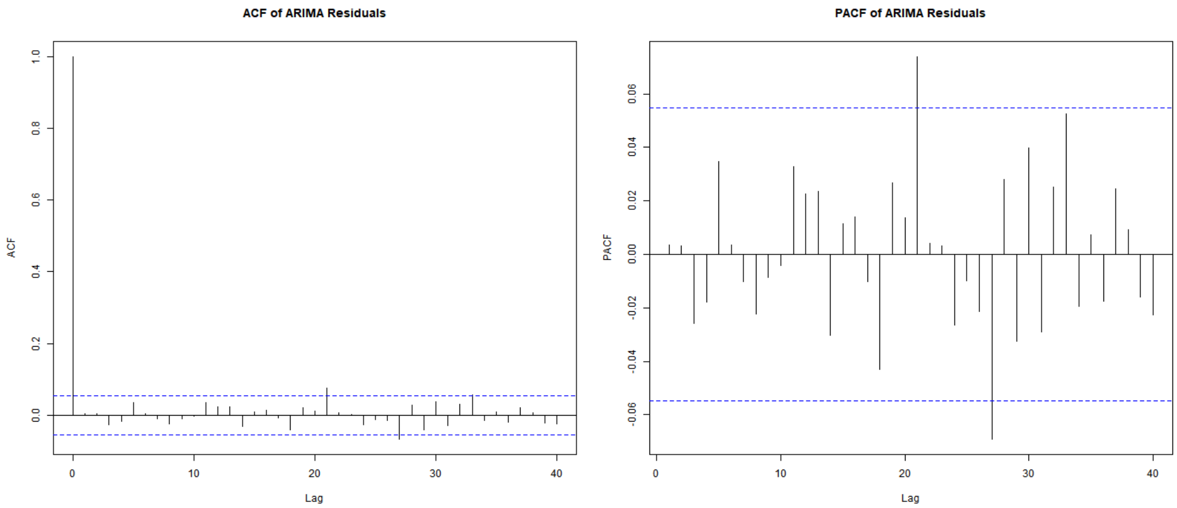

Figure 3: ARIMA residual ACF and PACF. The residuals show limited remaining autocorrelation, consistent with the Ljung-Box result.

The ARIMA residual checks are cleaner than the regression diagnostics. This is why the report does not simply choose the regression model as the best model overall. Regression was slightly better for point accuracy, while ARIMA was better behaved diagnostically.

7 Risk signal analysis

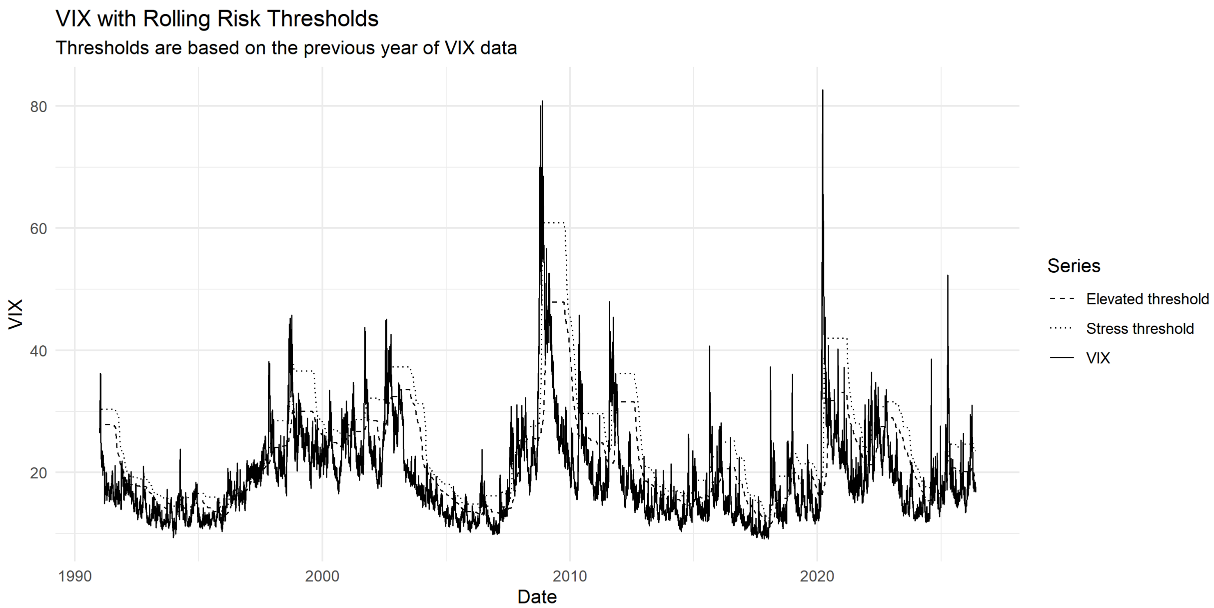

Figure 4: VIX with rolling risk thresholds. The elevated and stress thresholds move over time because they are based on the previous year of VIX data.

The rolling thresholds adapt to the volatility environment. This is useful because a VIX value that is high in a calm market may not be unusual during a persistent high-volatility period. The signal therefore measures risk relative to recent conditions, not against a fixed VIX level.

Table 5: Overall risk-regime summary

Regime	Days	Share (%)	Avg VIX	Median VIX	Max VIX	Avg z-score
Stress	1144	12.80	27.93	25.47	82.69	2.39
Elevated	1138	12.73	22.87	20.69	59.93	0.91
Normal	6657	74.47	17.28	16.11	46.35	-0.62

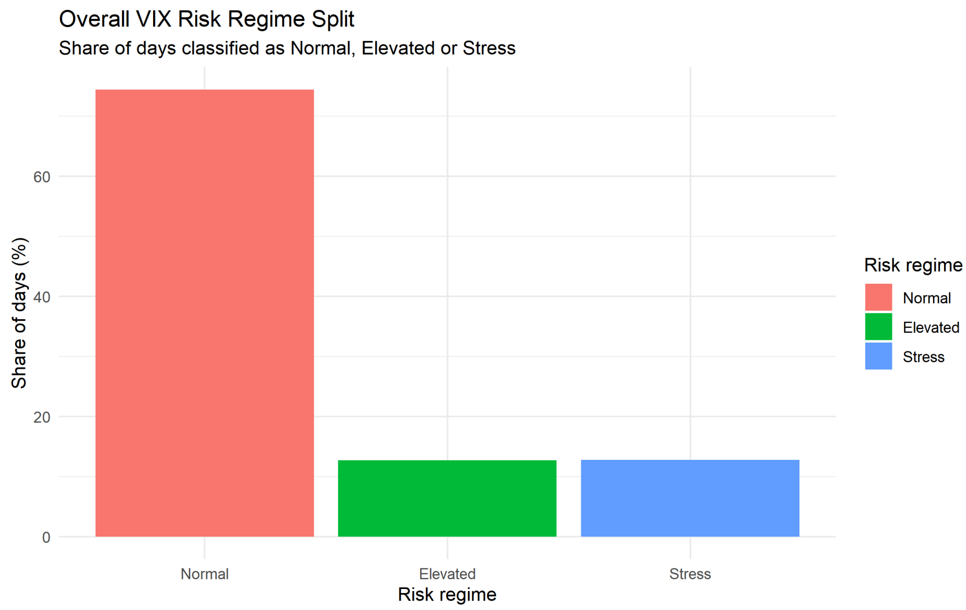

Figure 5: Overall split of Normal, Elevated and Stress regimes.

Normal periods make up most of the sample, while Elevated and Stress regimes are much smaller but have clearly higher average VIX values. The stress regime has an average VIX of 27.93 and includes the largest observed VIX spike in the sample.

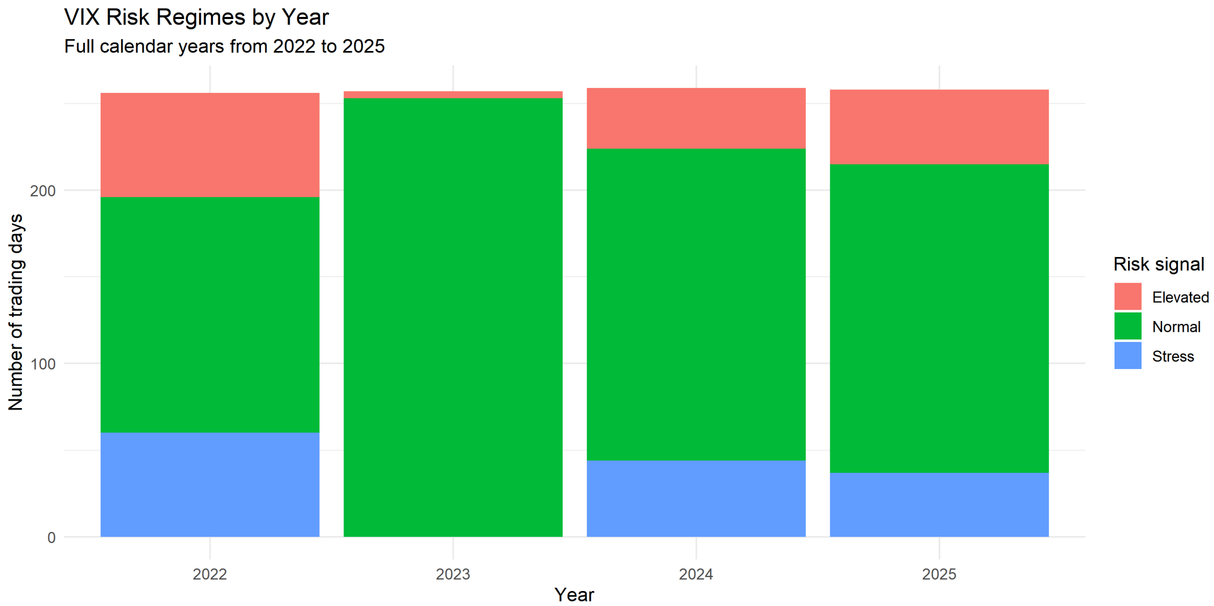

Figure 6: Risk regimes by year for full calendar years 2022 to 2025.

Table 6: Risk-regime days by year

Year	Normal	Elevated	Stress
2022	136	60	60
2023	253	4	0
2024	180	35	44
2025	178	43	37

The yearly view shows that 2022 had a large number of Elevated and Stress days, while 2023 was almost entirely Normal. Stress periods reappeared in 2024 and 2025. Because the thresholds are rolling, the regime label is relative to the recent volatility environment.

8 Latest signal

At the final date in the sample, the VIX was 16.76 and the signal was **Normal**. The elevated threshold was 19.19 and the stress threshold was 23.56. Since the observed VIX was below both thresholds, the final observation was classified as Normal.

Table 7: Latest risk signal

Date	VIX	Signal	z-score	Elevated threshold	Stress threshold
2026-05-21	16.76	Normal	-0.43	19.19	23.56

9 Conclusion

The project produces a coherent VIX forecasting and monitoring workflow. The regression model achieved the lowest RMSE in the rolling backtest, but the improvement over ARIMA was small. ARIMA produced cleaner residual diagnostics, so it remains a useful benchmark and a sensible model for the final forecast.

The risk signal is the most interpretable output from the project. It classifies VIX conditions into Normal, Elevated and Stress regimes using thresholds that are based only on past data. This gives a simple framework for monitoring market conditions without relying on fixed VIX cut-offs.

The main limitation is that the project forecasts the VIX index itself. VIX is an implied volatility measure, so the analysis should not be described as a direct realised-volatility forecast. A useful extension would be to compare VIX forecasts with realised S&P 500 volatility or to add a simple persistence benchmark.

References

[1] Federal Reserve Bank of St. Louis. *CBOE Volatility Index: VIX (VIXCLS)*. FRED economic data series. Available at: `https://fred.stlouisfed.org/series/VIXCLS`.

[2] Cboe Global Markets. *Cboe Volatility Index Methodology*. Cboe Indices methodology document. Available at: `https://cdn.cboe.com/resources/indices/Volatility_Index_Methodology_Cboe_Volatility_Index.pdf`.

[3] Hyndman, R. J. and Khandakar, Y. (2008). Automatic time series forecasting: the forecast package for R. *Journal of Statistical Software*, 27(3), 1-22.

[4] Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716-723.

[5] Ljung, G. M. and Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297-303.